HADI KHOSRAVINEZHAD

Electrical Engineering Graduate

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Academic Profile

Electrical Engineering graduate specializing in microwave photonics, high-speed electro-optic modulation, and high-frequency communication systems. First author of two published Q2 journal papers and one manuscript currently under review. Experienced in analytical model implementation, advanced multi-physics simulation using Ansys Lumerical, and MATLAB coding. Self-directed researcher with proven academic initiative in quantum computing concepts and educational curriculum development.

Education

Persian Gulf University — Bushehr, Iran
B.Sc. in Electrical Engineering — Sept 2021 – Aug 2026

- **Overall GPA:** 14.82 / 20 *(German Equivalent: ≈ 2.55 / Gut)*
- **Relevant Coursework:** Electromagnetic Theory, Fields and Waves, Microwave Engineering, Antenna Engineering, Communication Systems.

Eshraq High School — Borazjan, Iran
Diploma in Mathematics and Physics — Sept 2018 – June 2021

Scientific Publications

1. **Khosravinezhad, H.**, Keshavarz, H., & Mansorinezhad, M. (2025). "Tunable microwave photonic mixer based on cascaded QCSE electro-absorption modulators for efficient frequency conversion." *Optics Continuum*, 4(8), 1662–1675. [DOI: 10.1364/OPTCON.564528] **[Q2 Journal]**

2. **Khosravinezhad, H.**, & Keshavarz, H. (2026). "Generalized analytical framework for nonlinear intermodulation analysis and waveform reconstruction in high-speed EOM-MZI modulators." *Applied Optics*, 65(16), 5491–5499. [DOI: 10.1364/AO.595863] **[Q2 Journal]**

3. **Khosravinezhad, H.**, & Keshavarz, H. (2026). "Analytical Modeling and Dynamic Range Optimization of Cascaded Dual-Drive MZM Microwave Photonic Down-Converters." *Manuscript Under Review in Optica Publishing Group* (Submitted: September 29, 2026).

Research Experience

IoT Laboratory, ICT Research Institute, Persian Gulf University — Bushehr, Iran
Undergraduate Researcher — Aug 2024 – Present

- Conducted research on microwave photonic systems, high-speed electro-optic modulators (EOM-MZI), and cascaded EAM mixers in collaboration with Dr. Heidar Keshavarz.
- Executed comprehensive multi-physics simulations using **Ansys Lumerical** (INTERCONNECT, FDTD, MODE) to cross-validate analytical formulations against numerical benchmarks.
- Developed custom **MATLAB** algorithms for nonlinear distortion analysis (IMD2/IMD3), Jacobi–Anger expansions, and meta-heuristic optimization frameworks (ADE-NM, CPSO-NM).
- Prepared technical documentation and authored initial manuscript drafts for peer-reviewed journal submissions.

Teaching Experience & Academic Projects

Teaching Assistant (TA) – Electromagnetics Persian Gulf University
Department of Electrical Engineering Two Academic Semesters

- Instructed undergraduate students in problem-solving sessions, focusing on electromagnetic fields, and Maxwell's equations.
- Authored a structured **Electromagnetics Study Booklet** in Persian to simplify difficult concepts for Persian-speaking students facing a lack of intuitive reference materials. **[GitHub Repo]**

Independent Study in Quantum Information & Computing Self-Directed Project

- Autonomously studied quantum mechanics principles, quantum algorithms, and computational models.
- Developed a comprehensive educational booklet titled **Quantum Computation** in Persian, tailored for Persian-speaking readers. **[GitHub Repo]**

Technical Skills

- **Optical & EM Simulation Tools:** Ansys Lumerical (INTERCONNECT, FDTD, MODE), CST Studio Suite, Ansys Maxwell, Keysight ADS.
- **Scientific Computing & Programming:** MATLAB (Analytical modeling, optimization algorithms, data visualization), Numerical Methods.
- **Academic Documentation:** LaTeX, MS Office.

Specialized Training & Certifications

- **Ansys Lumerical Specialized Training:** Ansys Optics Official Learning Resources Feb 2024
- **Ansys Maxwell Simulation Training:** PowerEn Academy 2023
- **Industrial Electrical Systems Certification:** Technical & Vocational Training Center, Bushehr Aug 2023

Languages

- **English:** Advanced / C1 Level – **TOEFL iBT: 100 / 120** *(Test Date: June 2026)*
  - Subscores (Scale of 1.0 to 6.0): **Reading:** 5.0 / 6.0 | **Listening:** 6.0 / 6.0 | **Speaking:** 5.0 / 6.0 | **Writing:** 4.5 / 6.0
- **German:** Intermediate (B2 Level – Self-assessed)
- **Persian:** Native Language

Research Interests

- Microwave Photonics (MWP) & Linearized Frequency Conversion Systems
- High-Speed Electro-Optic Modulation (EOM) & Photonic Integrated Circuits (PICs)
- Intermodulation Distortion (IMD3) Cancellation & Dynamic Range Optimization
- Quantum Engineering, Quantum Optics & Quantum Information Processing

Personal Interests

- **Endurance Sports:** Running, Cycling, Hiking
- **Strategic Thinking:** Chess, Competitive Esports